ADE Power BI - Performance Improvements

Goal:

- Initial Goal: Complete the performance analysis process (interpret results) based on production data, to be able to form hypothesis and conclusions, which could inform and provide positive insight on current system performance and bottlenecks.

Objectives:

- Detect the bottlenecks to be tuned
- Assist the development team in determining the performance characteristics for various configuration options.

Test Scenarios:

#	Category	Proposed Change	Environment	Result	Recommendation	Work Item #	Notes	Status
1a	Capacity Management Settings	Limit the impact of expensive Power BI Queries on the **fabcapgeneralprod02** capacity by reducing the query timeout. Configure settings for query timeout - decrease default timeout to terminate long-running queries sooner. An integer that defines the timeout, in	FABCAPGEN...		IMPLEMENT		1. Power BI Reports: The default timeout is 225 seconds . The Capacity Query Timeout setting cannot override this if it is set to a higher value. The	NOT ST...

		seconds, for queries. The default is 3600 seconds (or 60 minutes). Note that Power BI reports will already override this default with a much smaller timeout for each of its queries to the capacity. Typically, it is approximately 3 minutes. **Setting Name:**Query Timeout (seconds) Verify Liam's suggested timeout value and determine what % of Queries would be impacted by viewing historical data.					lower timeout always takes precedence, meaning the Capacity Query Timeout only affects Power BI reports if set below 225 seconds. 2. SSAS and AAS: The same logic applies, but their default query timeout is 600 seconds. If the Capacity Query Timeout is set below 600 seconds, it will take effect. For example, setting 225 seconds in the Capacity settings will impact SSAS/AAS	

queries since the lower value applies.

3. External Tools (XMLA, Analyze in Excel, etc.): There is no default limit for these tools, so any value set in the Capacity Query Timeout will apply to them.
4. Workspace-Level Query Timeout: This must be configured via SQL Server Managemen t Studio (SSMS). Power BI tenant admins cannot enforce query timeout settings

							across users' workspaces (at workspace level) unless the admins have admin access to user's workspaces. See: Power BI Premium new capacity settings allow for more control over datasets | Blog di Microsoft Power BI | Microsoft Power BI	
1b	Capacity Management Settings	Configure settings for memory usage - set DAX query row returns to protect memory. Applies only to DAX measures and queries. Specified in % and limits how much memory can be used by temporary results during a query.	FABCAPGEN...		IMPLEMENT			NOT ST...

		Setting Name: Setting Name:Query Memory Limit (%)						
1c	Capacity Management Settings	Max Intermediate Row Set Count The max number of intermediate rows returned by DirectQuery. Default value is set to 1000000 and allowable range is between 100000 and 2147483647 **Setting Name:** Max Intermediate Row Set Count	FABCAPGEN...		IMPLEMENT			NOT ST...
1d	Capacity Management Settings	Sometimes, a user can execute an expensive DAX query that returns a very large number of rows. This can cause a lot of resource usage and affect other users and report capacity administrator to limit how many rows should be returned for any individual DAX query. The max number of intermediate rows returned by DirectQuery. Default value is set to	FABCAPGEN...		IMPLEMENT			NOT ST...

		1000000 and allowable range is between 100000 and 2147483647 **Setting Name:** Max Result Row Set Count						
1e	Capacity Management Settings	When suers are experiencing slowness due to a large dataset taking up memory resources , admins would very often end up in the similar cycle of first identifying the culprit datasets, contacting the owner or migrating to a different capacity. With this new setting, admins can now configure the dataset size and prevent report creators from publishing a large dataset that could potentially take down the capacity and secondly save the admin from the painful cycle of identifying and mitigating. Maximum size of the offline dataset in	FABCAPGEN...		ON-HOLD		Look for evidence of this scenario	NOT ST...

		memory. This is the compressed size on disk. Default value is set by SKU and the allowable range is from 0.1 – 10 GB **Setting Name:** Max Offline Dataset Size (GB)						
4	User Training	Identify users with poor interactive queries and guide them on best practices	FABCAPGEN...		IMPLEMENT		Work out who the target users are and gather evidence that they require training Review the executed DAX queries initiated by the user. See ADE Realtime monitoring dashboard for queries sorted by CPU time descending ADE Realtime Workspace Monitoring Dashboard Top 10 Users by MAX CPU Time	NOT ST...
5	Capacity	Workload management, move	FABCAPGEN...		IMPLEMENT		Need to check whether any existing	COMP...

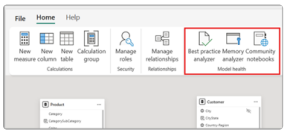
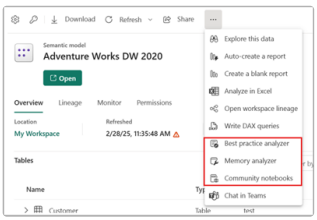
	Management	any non-Prod workspaces from **fabcapgeneralprod02** to another fabric capacity. DEV and E2E workspaces are currently hosted in FABCAPGENERALPROD02					CI/CD pipelines are impacted i.e assume workspaces were manually created and that redeploying artefacts will not impact any workspaces that have moved.	
5a	Surge Protection	Introduce **Capacity Level** Surge Protection Measures The Background operations rejection threshold determines when surge protection becomes active. The threshold is compared to the capacity's 24-hour background percentage, representing the smoothed average committed utilization projected over the next 24 hours. When the threshold is reached or exceeded, surge protection becomes active. When surge protection is active, the capacity rejects new background operations. When surge protection isn't	FABCAPGEN...		ON-HOLD		Requires testing and monitoring Open the Fabric Admin Portal. Navigate to Capacity settings. Select FABCAPGENERALPROD02 Expand Surge protection. Set Background Operations to On. Set a Rejection threshold. Set a Recovery threshold. Select Apply. To monitor surge protection, follow these steps: Open the Microsoft Fabric Capacity Metrics app. On the Compute page, select System	NOT ST...

		enabled, the 24-hour background percentage is allowed to reach 100% before the capacity rejects new background operations. The Background operations recovery threshold determines when surge protection stops being active. Surge protection stops being active when the 24-hour background percentage drops below the background recovery threshold you set. At this point, the capacity starts to accept new background operations. • **Limit background compute consumption** at the capacity level, setting thresholds for when background operations are rejected and when they recover. • **Reduce deep throttling states with longer recover**					events. The System events table shows when surge protection became active and when the capacity returned to a not overloaded state. You can also find this information from the States table within Real Time Hub capacity events.	

		time by engaging throttling earlier to enable admins to take preventive action and reducing recovery times. **Setting Name:** Background operations rejection threshold Background operations recovery threshold						
5b	Surge Protection	Introduce **Workspace Level** Surge Protection Measures CU consumption limits per workspace: Define maximum CU consumption limits per workspaces within a capacity for a rolling 24 hour window. The workspace CU limit is set as a percentage threshold that applies to all workspaces in the capacity unless a workspace is marked mission critical or blocked. Per workspace-level configurations:	FABCAPGEN...		ON-HOLD			NOT ST...

		Configure surge protection settings for each workspace with one of the following states: **Available:** Workspace follows the capacity-level surge protection rules. **Mission Critical:** Workspace ignores all capacity-level surge protection rules (immune from blocking). **Blocked:** Workspace is manually or automatically blocked for a specified period; rejecting all operation requests. ADE - Trade candidate for mission critical exclusion						
6a	Semantic Model Optimisation	Investigate and optimize the semantic model and its measure definitions, etc. Explore Semantic Model Meta Data Analyser [Explore SMMDA]()	FABCAPGEN...		INPLEMENT		Decide which semantic models warrant further investigation. 1. **Navigate** to the `FUAM_Semantic_Model_Meta_Data_Analyzer_Report` item	IN PRO...

							within the workspace where FUAM has been deployed (see [FUAM Workspace](#)) 2. **Open** the [report.](#) 3. **Navigate** to the "How to configure" report page. 4. **Follow** the steps described on that report page. 5. **Open** the `FUAM_Semantic_Model_Meta_Data_Analyzer_Report` again or **refresh** the web browser page with the opened report. Assign to Power Bees for further investigatio n.	
6b	Semantic Model Optimisation	Investigate and optimize the semantic model and its measure definitions, etc. **Open sample notebooks from the semantic model**	DEV		ON-HOLD		Decide which semantic models warrant further investigation. Power Bees team already using BPA with Tabular Editory	NOT ST...

		Notebooks offer a collection of ready-made experiences for semantic models to help you get started, which you can access in multiple ways: • From the **Home** tab of the ribbon, editing semantic models in the Power BI service, shown in the following image.  • From the semantic model details page under the **Model health** dropdown, shown in the following image. • From **More options (...)** in the OneLake catalog when selecting a semantic model, shown in the following image. 					Use Fabric notebooks and semantic link, to analyze semantic models	

		Explore the Best Practices Analyser Learn more						
6c	Semantic Model Optimisation	Investigate and optimize the semantic model and its measure definitions, etc. When you run the sample notebook notebook, the **Memory Analyzer** shows you memory and storage statistics about the objects in your semantic model, such as Tables, Columns, Hierarchies, Partitions, and Relationships. The statistics the Memory Analyzer returns can be used to identify areas of performance optimization and memory reduction for your semantic model. The following image shows the Memory Analyzer in use. Explore the Memory Analyser Learn more	DEV		ON-HOLD		Decide which semantic models warrant further investigation.	NOT ST...
6d	Semantic	Use Bravo to quickly analyze where your	DEV		ON-HOLD		Decide which semantic models	NOT ST...

	Model Optimisation	model consumes the most memory and choose which columns to remove to optimize it. Bravo for Power BI by SQLBI					warrant further investigation.	
6e	Semantic Model Optimisation	Use the Fabric Load test tool for performance testing Microsoft Fabric semantic models under concurrent load conditions. Simulates multiple users executing DAX queries simultaneously against your semantic models fabric-toolbox/tools/FabricLoadTestTool at main · microsoft/fabric-toolbox	DEV		ON-HOLD		Decide which semantic models warrant further investigation. Fabric Load Test tool deployed here: Fabric Load Test Tool Workspace	NOT ST... NOT ST...
6f	Semantic Model Memory Optimisation	TODO Optimizing Semantic Model Memory in Fabric	DEV		ON-HOLD			ON-HO...
7a	Report Optimisation	Use Power BI Inspector GitHub - NatVanG/PBI-InspectorV2: Rules	DEV		ON-HOLD		Decide which reports warrant further investigation. User feedback	NOT ST...

		-based metadata testing for the Microsoft Power BI visual layer and other Fabric Items.					required, or when refresh is slow or errors upon refresh,	
7b	Report Optimisation	Use Measure Killer Optimise and Clean Power BI Reports Measure Killer optimization tool	DEV		ON-HOLD		Decide which reports warrant further investigation. User feedback required, or when refresh is slow or errors upon refresh,	NOT ST...
8a	DAX Performance	Use the DAX Performance testing notebook to measure DAX query timings under different cache states (cold, warm, and hot). fabric-toolbox/tools/DAXPerformanceTesting at main · microsoft/fabric-toolbox	DEV		ON-HOLD		Decide which reports warrant further investigation. Notebook deployed here: DAX Performance Testing Workspace	NOT ST...
8b	DAX Performance	Use the DAX Performance Tuner MCP Server to optimise DAX queries fabric-toolbox/tools/DAXPerformanceTunerMCPServer at main · microsoft/fabric-toolbox	DEV		ON-HOLD		Decide which reports warrant further investigation.	NOT ST...

9	Capacity Overage	Enable Capacity Overage (preview feature)	FABCAPGEN...		ON-HOLD		Introducing capacity overage: Flexibility when you need it most (Preview) | Microsoft Fabric-Blog | Microsoft Fabric	NOT ST...
10	Query Scale Out	Semantic model scale-out uses your Premium capacity to host one or more read-only replicas of your primary semantic model. By increasing throughput, the read-only replicas ensure performance doesnâ€™t slow down when multiple users submit queries at the same time.	FABCAPGEN...		ON-HOLD		See Power BI refresh, memory usage and Semantic Model Scale Out Power BI semantic model scale-out - Microsoft Fabric semantic-link-labs/notebooks/Query Scale Out.ipynb at main · microsoft/semantic-link-labs	NOT ST...
11	Self-serve on a different capacity	Official Power BI reports built by the Power BI Engineering team require consistent and stable performance. These reports are typically designed and tested with performance in mind, for example by controlling filters (default time periods, preselections etc)..	FABCAPGEN...		ON-HOLD			NOT ST...

limiting data volumes, and following best practices.

In contrast, self-serve reporting introduces a level of unpredictability. Users may combine large volumes of data, apply complex or unoptimised filters, use a high number of heavy visuals, or not follow recommended design guidance.

These can cause spikes which negatively impact the performance of official reports deployed by the Power Bi team

Can we introduce a separate, smaller capacity dedicated to self-serve reporting?

This would allow us to:

- Isolate performance impact from self-serve workloads
- Protect the stability of enterprise reports
- Identify and address inefficient or high-impact self-serve reports early (Why is

		the user doing this, cant they use X report instead etc.) • Better understand emerging reporting needs before promoting them to enterprise-level solutions This approach would enable us to support self-serve flexibility while maintaining performance and reliability for core reporting.						

Links:
ADE Monitoring Dashboard:
[ADE Real-time Monitoring Dashboard](#) (Real-time dashboard)
[Fabric Monitoring Workspace Report](#) (Semantic Model - Performance Monitoring)